

Notes: Begenau and Landvoigt (RES, 2022)

Financial Regulation in a Quantitative Model of the Modern Banking System

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1 Big picture

- **Question.** What happens when regulators tighten capital requirements on commercial banks in an economy where unregulated shadow banks can also provide credit and money-like liabilities? Does activity simply leak into the shadow banking sector, making the system more fragile?
- **Core issue.** Capital regulation is usually evaluated through the regulated banking sector. This paper asks for the general equilibrium answer: regulated banks, unregulated banks, households, liquidity premia, deposit insurance, default, and runs all adjust jointly.
- **Setting.** The model has two bank types:
 - **C-banks:** regulated commercial banks. Their debt is insured, so depositors view it as safe. They face a capital requirement.
 - **S-banks:** unregulated shadow banks. Their debt is risky, may be bailed out only probabilistically, and can be exposed to run shocks.
- **Main theoretical force 1: competition effect.** Deposit insurance gives C-banks a competitive advantage. They can issue safe, liquidity-producing debt at favorable prices. This crowds out S-banks and makes S-bank liquidity relatively scarce. Because S-bank debt then earns a high convenience yield, S-banks lever up to compete for equity capital. Raising C-bank capital requirements weakens the C-bank advantage and reduces this competitive pressure on S-banks.
- **Main theoretical force 2: demand effect.** Households have downward-sloping demand for aggregate liquidity. When C-banks are forced to issue less debt, aggregate liquidity supply falls and convenience yields rise. Higher convenience yields make debt financing more attractive for S-banks and push them toward higher leverage.
- **Central ambiguity.** A tighter C-bank capital requirement always reduces C-bank leverage and shifts some activity toward S-banks. But the effect on S-bank leverage is ambiguous: the competition effect lowers S-bank risk-taking, while the demand effect raises it.
- **Quantitative answer.** In the calibrated U.S. model, the demand effect dominates for S-bank leverage. S-banks become larger and slightly riskier after tighter C-bank requirements. However, the large fall in C-bank risk dominates the increase in S-bank risk.
- **Main policy result.** The calibrated optimal C-bank capital requirement is about **16%**. Moving from the benchmark capital requirement to 16% raises welfare by about **0.054%** in consumption-equivalent terms in the paper's calculation.
- **Important nuance.** More shadow banking does not automatically mean a riskier financial system. In this model, tighter regulation creates more shadow banking but a safer overall financial system.
- **Post-crisis interpretation.** The model can rationalize some post-2008 patterns. Higher C-bank requirements alone would raise the shadow-bank share, but post-crisis reductions in implicit bailout expectations and the recognition of run risk push the shadow-bank share down.
- **Economic takeaway.** Regulation has leakage, but leakage is not the whole story. The relevant object is the equilibrium interaction between deposit insurance, liquidity demand, and competition for household equity capital.

2 Data and measurement (key objects)

2.1 The quantitative target: a modern banking system

- **Sample period.** The calibration uses quarterly U.S. data from **1999Q1–2019Q4**. The start date is chosen because 1999 marks the Gramm–Leach–Bliley deregulation period, which makes the bank holding company definition more stable.
- **Data sources.** The paper combines bank holding company call reports, Compustat/CRSP, Flow of Funds, NIPA, FRED, Moody’s default and recovery data, and estimates from related papers on shadow banking and liquidity premia.
- **Unit of analysis.** The model is not an empirical regression paper. It is a calibrated and simulated general equilibrium model. The empirical work is used to discipline parameters and validate model moments.
- **Bank-dependent sector.** Bank-dependent production is mapped to output from firms without S&P long-term credit ratings, plus real estate and construction-related industries because mortgage-related lending is central to bank balance sheets.
- **Commercial banks.** C-banks are mapped to publicly traded depository institutions and bank holding companies.
- **Shadow banks.** S-banks are mapped to finance companies, GSE-related firms, REITs, and miscellaneous investment firms. The paper’s definition is liability-oriented: the key issue is money-like short-term liabilities, not simply the type of asset held.

Interpretation. The quantitative exercise is designed to ask: if the U.S. financial system has both insured deposit-taking banks and uninsured money-market-like intermediaries, what is the welfare-maximizing regulation of insured banks?

2.2 Liquidity objects

Households value the liabilities of both bank types because they provide liquidity services. The key quantities are:

- A_t^C : household holdings of C-bank deposits;
- A_t^S : household holdings of S-bank debt;
- $H(A_t^S, A_t^C)$: aggregate liquidity services;
- C-bank convenience yield: the marginal liquidity value of insured deposits;
- S-bank convenience yield: the marginal liquidity value of risky shadow-bank debt.

Interpretation. The model treats both bank liabilities as money-like, but not as perfect substitutes. Insured deposits are safer and receive a larger calibrated liquidity weight.

2.3 Risk objects

- **C-bank default.** C-bank default is borne by the government through deposit insurance. Depositors are paid in full, but default creates real resource losses and fiscal costs.

- **S-bank default.** S-bank default is borne by depositors unless a bailout occurs. Bailouts happen with probability π_B rather than as full insurance.
- **Runs.** A run is modeled as an early withdrawal shock. A fraction π_t^R of S-bank deposits is withdrawn early, forcing S-banks to liquidate assets at a low household productivity value.
- **Bankruptcy losses.** Default destroys resources through bankruptcy and liquidation costs. These costs are calibrated separately for C-banks and S-banks using recovery data.

Interpretation. The welfare trade-off is not simply “more debt versus less debt.” Debt creates liquidity services but also default and run fragility.

2.4 Key calibrated moments

The most important quantitative anchors are:

- benchmark C-bank capital requirement: about **10%**;
- C-bank deposit interest rate: about **0.36% per quarter** in the data;
- S-bank share of liquidity funding: about **34%**;
- option-implied C-bank liquidity premium target: about **21 bps per quarter**;
- S-bank leverage target: about **87%** in the data, **83.2%** in the model;
- C-bank default/charge-off target: about **0.23%**;
- S-bank bond default target: about **0.28%**;
- run probability: about **4.76%** unconditionally;
- average run length: about **1.3 quarters**.

3 Model and mechanisms (all equations + interpretation)

3.1 Household preferences and liquidity services

The representative household values consumption and liquidity services:

$$U(C_t, H(A_t^S, A_t^C)) = \frac{C_t^{1-\gamma}}{1-\gamma} + \psi \frac{\left[\alpha (A_t^S)^\varepsilon + (1-\alpha)(A_t^C)^\varepsilon \right]^{\frac{1-\gamma_H}{\varepsilon}}}{1-\gamma_H}. \quad (2.1)$$

Objects.

- γ : inverse intertemporal elasticity of substitution for consumption.
- ψ : weight on liquidity services.
- α : weight on S-bank liquidity.
- ε : governs substitutability between C-bank and S-bank liquidity.

- γ_H : curvature in aggregate liquidity services; when $\gamma_H > 0$, liquidity demand is downward sloping.

Interpretation. Households want bank liabilities not only for pecuniary payoffs but also because they are money-like. The CES structure means insured deposits and shadow-bank debt are imperfect substitutes. This imperfection is essential: regulation changes not only the quantity of liquidity but also its composition.

3.2 Bank production technology

Both C-banks and S-banks operate the same bank-dependent production technology. For bank type $j \in \{C, S\}$,

$$Y_t^j = Z_t(\widehat{K}_t^j)^{1-\eta}(N_t^j)^\eta. \quad (1)$$

After substituting optimal labor and investment, the payoff per unit of bank capital is

$$\Phi_t^j = (1 - \eta)Z_t(n_t^j)^\eta + p_t - \delta_K + \frac{(p_t - 1)^2}{2\phi_I}. \quad (2.2)$$

Households can also operate capital, but less efficiently:

$$\Phi_t^H = (1 - \eta)\bar{Z}_t(n_t^H)^\eta + p_t(1 - \bar{\delta}_K). \quad (2.3)$$

Interpretation. Both bank types have the same asset-side technology. This is deliberate: the paper isolates liability-side differences—deposit insurance, capital requirements, bailouts, and run exposure—rather than giving shadow banks different assets.

3.3 S-bank runs

A fraction π_t^R of S-bank deposits may be withdrawn early:

$$\pi_t^R \in \{0, \bar{\pi}^R\}. \quad (2)$$

If a run occurs, S-banks must liquidate a fraction of assets:

$$\Delta_t^S = \frac{\pi_t^R B_t^S}{K_t^S \Phi_t^H}. \quad (3)$$

The capital left for production is

$$\widehat{K}_t^S = (1 - \Delta_t^S)K_t^S. \quad (4)$$

Interpretation. Runs are costly because S-banks must sell assets to households who are inefficient operators of capital. The run destroys value even before formal default because it forces low-productivity liquidation.

3.4 S-bank portfolio problem

S-banks choose next-period capital and debt. Let

$$b_{t+1}^S = \frac{B_{t+1}^S}{K_{t+1}^S}, \quad k_{t+1}^S = \frac{K_{t+1}^S}{\widehat{K}_t^S}. \quad (5)$$

The paper rewrites the S-bank problem per unit of capital as

$$v_t^S = \max_{b_{t+1}^S \geq 0, k_{t+1}^S \geq 0} -k_{t+1}^S \left[p_t - q_t^S(b_{t+1}^S)b_{t+1}^S + \frac{\phi_K}{2}(k_{t+1}^S - 1)^2 \right] \\ + k_{t+1}^S \mathbb{E}_t \left[M_{t,t+1} \Phi_{t+1}^S \max \left\{ (1 - \Delta_{t+1}^S) \rho_{t+1}^S + \frac{v_{t+1}^S}{\Phi_{t+1}^S} - L_{t+1}^S (1 - \pi_{t+1}^R), -\delta_S (1 - \Delta_{t+1}^S) \right\} \right]. \quad (2.7)$$

Interpretation.

- The first line is the cost of building next period's balance sheet.
- The second line is the expected discounted value of future equity.
- The max operator captures limited liability and the option to default.
- The price of S-bank debt depends on S-bank leverage because households know S-bank debt is risky.

The S-bank default threshold is

$$\widehat{\rho}_t^S = \frac{(1 - \pi_t^R) L_t^S - (1 - \Delta_t^S)(v_t^S / \Phi_t^S + \delta_S)}{1 - \Delta_t^S}. \quad (2.8)$$

Interpretation. S-banks default when the idiosyncratic payoff shock is sufficiently low. Higher leverage or larger run-induced withdrawals raise the default threshold.

3.5 C-bank problem and capital regulation

C-banks differ from S-banks in four ways: their deposits are insured, they do not suffer runs, they face a capital requirement, and they pay a deposit insurance fee. Their problem is

$$v_t^C = \max_{b_{t+1}^C \geq 0, k_{t+1}^C \geq 0} -k_{t+1}^C \left[p_t - (q_t^C - \kappa) b_{t+1}^C + \frac{\phi_K}{2}(k_{t+1}^C - 1)^2 \right] \\ + \mathbb{E}_t \left[M_{t,t+1} k_{t+1}^C \Phi_{t+1}^C \max \left\{ \rho_{t+1}^C + \frac{v_{t+1}^C}{\Phi_{t+1}^C} - L_{t+1}^C, -\delta_C \right\} \right], \quad (2.9)$$

subject to

$$(1 - \theta) p_t \geq b_{t+1}^C. \quad (2.10)$$

The C-bank default cutoff is

$$\widehat{\rho}_t^C = L_t^C - \frac{v_t^C}{\Phi_t^C} - \delta_C. \quad (2.11)$$

Interpretation. The capital requirement directly limits insured debt issuance. Because C-bank debt is insured, C-banks do not internalize depositor losses in the same way as S-banks. This is why deposit insurance creates a competitive advantage and a distortion.

3.6 Government guarantees and fiscal costs

If a bank defaults, its assets are liquidated and some resources are lost. The recovery value per unit of debt is

$$r_t^j = (1 - \xi_j) \frac{\rho_t^{j,-} (1 - \Delta_t^j)}{L_t^j (1 - \pi_t^R \mathbf{1}_{j=S})}, \quad j \in \{C, S\}. \quad (6)$$

Lump-sum taxes finance C-bank deposit insurance and probabilistic S-bank bailouts:

$$T_t = F_{\rho,t}^C (1 - r_t^C) B_t^C - \kappa B_{t+1}^C + \pi_B F_{\rho,t}^S (1 - r_t^S) (1 - \pi_t^R) B_t^S. \quad (7)$$

Interpretation. C-bank guarantees are automatic. S-bank bailouts are probabilistic. Both affect equilibrium because they change debt prices, bank leverage, and ultimately the amount of liquidity produced by each sector.

3.7 Household budget constraint and equilibrium

Households buy bank equity and bank debt:

$$W_t + Y_t + w_t - T_t \geq C_t + \sum_{j \in \{C, S\}} p_t^j S_t^j + \sum_{j \in \{C, S\}} q_t^j A_{t+1}^j. \quad (2.12)$$

Market clearing requires

$$A_{t+1}^j = B_{t+1}^j, \quad S_t^j = 1, \quad N_t^C + N_t^S + N_t^H = 1. \quad (8)$$

Interpretation. All debt and equity issued by banks must be held by households. This is why competition for household capital matters: C-banks and S-banks must offer the same equilibrium risk-adjusted return on equity.

4 Simplified two-period model (mechanism and propositions)

4.1 Why the paper uses a simple model

Before the full calibration, the paper strips the model to two periods to show analytically why a higher C-bank capital requirement can both expand shadow banking and improve welfare.

The simplified household utility is

$$U = C_0 + \beta \left(C_1 + \psi H(A^S, A^C) \right). \quad (3.13)$$

Interpretation. The simple model isolates liquidity and default. There is no dynamic simulation yet, but the same economic trade-off is present: bank debt creates liquidity and default risk.

4.2 S-bank and C-bank problems

S-banks solve

$$\max_{K^S \geq 0, B^S \geq 0} q^S(B^S, K^S) B^S - p K^S + \beta \mathbb{E} \left[\max\{\rho^S K^S - B^S, 0\} \right]. \quad (3.14)$$

C-banks solve

$$\max_{K^C \geq 0, B^C \geq 0} q^C B^C - pK^C + \beta \mathbb{E} \left[\max\{\rho^C K^C - B^C, 0\} \right] \quad (3.15)$$

subject to

$$B^C \leq (1 - \theta) \mathbb{E}(\rho^C K^C). \quad (3.16)$$

Interpretation. S-bank debt is priced by its default risk. C-bank debt is protected by deposit insurance, so its price does not fall one-for-one with bank risk. This asymmetry is the source of the C-bank advantage.

4.3 Planner allocation

The planner chooses S-bank capital share and leverage to maximize consumption plus liquidity:

$$\max_{K^S, L^S, L^C} K^S(1 - F_S(L^S))\rho_S^+ + (1 - K^S)(1 - F_C(L^C))\rho_C^+ + \psi H(L^S K^S, L^C(1 - K^S)). \quad (3.20)$$

If the two bank types have the same idiosyncratic shock distribution, the efficient allocation satisfies

$$A^* \equiv \frac{A^S}{A^C} = \frac{K^S}{K^C} = \left(\frac{\alpha}{1 - \alpha} \right)^{\frac{1}{1 - \varepsilon}}, \quad (3.21)$$

and leverage is equalized across bank types.

Interpretation. The planner allocates liquidity production according to the relative usefulness of each type of liquidity. If S-bank and C-bank technologies are symmetric, the planner does not want one type to be much more levered than the other.

4.4 Debt pricing equations

Households price C-bank and S-bank debt according to

$$q^C = \beta \left(1 + \psi H_C(A^S, A^C) \right), \quad (3.22)$$

$$q^S = \beta \left(1 - F_S(L^S) + \psi H_S(A^S, A^C) \right). \quad (3.23)$$

Interpretation.

- C-bank debt pays one for sure because of deposit insurance.
- S-bank debt has expected repayment $1 - F_S(L^S)$ because it can default.
- Both debts carry a liquidity premium, but their premia depend on the equilibrium quantities of both debt types.

4.5 Capital market condition and the competition effect

Because bank equity markets are competitive, expected equity values must be equalized across bank types. This gives the key capital market condition:

$$\begin{aligned} & (1 - F_S(L^S))\rho_S^+ - (1 - F_C(L^C))\rho_C^+ + \psi \left(L^S H_S(A^S, A^C) - L^C H_C(A^S, A^C) \right) \\ & = F_C(L^C) L^C. \end{aligned} \quad (3.26)$$

Interpretation. The right-hand side is the extra benefit C-banks get from insured deposits. S-banks must compensate investors either through higher asset payoffs or through higher liquidity premia. Since assets are symmetric in the benchmark theory, the adjustment works mainly through the scarcity and convenience yield of S-bank debt.

4.6 Distortion relative to the planner

The paper defines a liquidity wedge m by

$$L^C f_C(L^C) = (1 + m) \psi H_C(A^S, A^C). \quad (3.27)$$

Interpretation. The wedge compares the marginal social cost of producing C-bank liquidity through more leverage to the marginal social benefit of C-bank liquidity. If $m > 0$, C-banks overproduce liquidity relative to its social value.

Under a tractable uniform-shock assumption, the S-bank shares in the competitive equilibrium satisfy

$$\frac{A^S}{A^C} = \left(\frac{1}{M} \right)^{\frac{1}{1-\varepsilon}} A^*, \quad \frac{K^S}{K^C} = (1 + m) \left(\frac{1}{M} \right)^{\frac{2-\varepsilon}{1-\varepsilon}} A^*, \quad (3.28)$$

where $M = \sqrt{(1 + m)(3 + m)}$.

Interpretation. The S-bank sector is too small relative to the planner allocation because C-banks enjoy insured-debt subsidies. This scarcity raises S-bank liquidity premia and pushes S-bank leverage above the planner level.

4.7 Effect of higher C-bank capital requirements

The paper's Proposition 3 gives the central comparative statics:

- a higher θ reduces C-bank leverage;
- a higher θ increases the S-bank share of debt and capital;
- the effect on S-bank leverage is ambiguous;
- if $m \geq 0$, a marginal increase in the C-bank capital requirement improves welfare.

The welfare derivative can be written as

$$\frac{dU(\theta)}{d\theta} = m \mathbb{E}(\rho^C) \psi H_C(A^S, A^C) K^C + \frac{dK^S}{d\theta} F_C(L^C) L^C. \quad (3.29)$$

Interpretation.

- The first term is the standard benefit of reducing excessive C-bank liquidity production.
- The second term is specific to this paper: tighter C-bank regulation expands the S-bank sector, correcting the composition distortion created by deposit insurance.
- A regulator who ignores S-banks misses this second term.

4.8 Competition effect versus demand effect

- **Competition effect.** Higher θ weakens C-banks. S-banks can expand without needing such high convenience yields. This tends to reduce S-bank leverage.
- **Demand effect.** Higher θ reduces aggregate liquidity supplied by C-banks. Because household liquidity demand is downward sloping when $\gamma_H > 0$, liquidity premia rise. This makes debt issuance more attractive and tends to increase S-bank leverage.
- **Special case.** If $\gamma_H = 0$, aggregate liquidity demand is perfectly elastic. The demand effect disappears, and higher C-bank capital requirements reduce S-bank leverage.
- **Quantitative result.** In the calibrated full model, $\gamma_H = 1.6$, so the demand effect is active and dominates for S-bank leverage.

5 Calibration and quantitative strategy (what makes the estimates credible)

5.1 Stochastic environment

The non-bank endowment follows

$$\log(Y_{t+1}) = (1 - \rho_Y) \log(\mu_Y) + \rho_Y \log(Y_t) + \epsilon_{t+1}^Y. \quad (9)$$

Bank-dependent productivity is

$$Z_t = \nu_Z Y_t \exp(\epsilon_t^Z). \quad (10)$$

Interpretation. Bank-dependent productivity inherits aggregate risk from GDP and adds bank-sector-specific risk. This lets the model generate both macro fluctuations and financial-sector fragility.

5.2 Solution method

- The model is solved globally as a nonlinear dynamic system.
- Unknown policy functions and prices are approximated with splines.
- The exogenous state variables are Y_t , Z_t , and the run shock π_t^R .
- The endogenous state variables are total bank capital, C-bank debt, S-bank debt, and the S-bank share of capital.
- The authors simulate the solved model for many periods and compare simulated moments to data moments.

Interpretation. The model is nonlinear because default, runs, capital constraints, and liquidity premia are all state-dependent. Local linearization would miss much of the policy-relevant mechanism.

5.3 Parameter groups

The calibration separates parameters into two groups:

- **Directly mapped parameters.** These have close one-to-one targets, such as depreciation, labor share, the benchmark capital requirement, run probability, and deposit insurance fees.
- **Jointly calibrated parameters.** These determine multiple moments together, such as bank output share, bank output volatility, default rates, recoveries, bailout probability, liquidity weight, and liquidity substitutability.

Interpretation. The paper does not claim each structural parameter is identified by a single regression. Instead, the model is disciplined by a system of moments that jointly pin down technologies, risks, liquidity demand, and regulation.

5.4 Liquidity-demand calibration

The most important calibration step estimates how relative debt prices respond to quantities of C-bank and S-bank liquidity. The debt price spread is approximately

$$q_t^C - q_t^S = \mathbb{E}_t \left[M_{t,t+1} \left(\text{MRS}_{t+1}^C - \text{MRS}_{t+1}^S + F_{\rho,t+1}^S \right) \right]. \quad (4.30)$$

The log-linearized equation motivates a regression of the C-bank/S-bank debt price spread on quantities of shadow-bank and commercial-bank debt:

$$\hat{q}_t^C - \hat{q}_t^S = \mathbb{E}_t \left[\bar{m} \widehat{M}_{t+1} + \bar{c} \widehat{C}_{t+1} + \beta_S \widehat{A}_{t+1}^S + \beta_C \widehat{A}_{t+1}^C + \bar{f} \widehat{F}_{\rho,t+1}^S \right]. \quad (4.31)$$

Read. In the data, the coefficient on log S-bank debt/GDP is about **-0.19%**, and the coefficient on log C-bank debt/GDP is about **0.50%**. The calibration targets these coefficients and obtains $\gamma_H = 1.6$ and $\varepsilon = 0.2$.

Interpretation. These regressions discipline the demand effect. They say that changes in the relative supply of C-bank and S-bank debt move relative liquidity premia, implying that the two liabilities are substitutes but not perfect substitutes.

5.5 Model fit

The model matches targeted moments reasonably well:

- bank-dependent output share: **22.3%** in both data and model;
- bank output growth volatility: **2.93%** in data and **3.02%** in model;
- C-bank default/charge-off rate: **0.23%** in both data and model;
- S-bank bond default: **0.28%** in data and **0.30%** in model;
- unsecured S-bank debt recovery: about **38.1%** in data and **38.2%** in model;
- C-bank recovery/resolution target: about **48.1%** in both data and model.

The model also produces reasonable untargeted patterns:

- consumption and investment are procyclical;
- C-bank book leverage is procyclical and C-bank market leverage is countercyclical;
- liquidity benefits are countercyclical because recessions reduce safe/liquid asset supply;
- the C-bank/S-bank rate spread has the same business-cycle sign as in the data.

Main limitation. The model generates too little C-bank leverage volatility because the C-bank capital constraint binds tightly. In the data, banks hold capital buffers and adjust leverage more flexibly.

6 Tables and figures

6.1 Table 1: Calibration

Read. Table 1 lists the structural parameters and the data moments used to discipline them. Panel A contains direct mappings; Panel B contains jointly matched moments.

Key numbers.

- $\theta = 10\%$: benchmark capital requirement.
- $\kappa = 0.142\%$: deposit insurance fee.
- $\alpha = 0.33$: S-bank weight in liquidity services.
- $\psi = 0.0072$: liquidity preference weight.
- $\gamma_H = 1.6$: curvature in liquidity demand.
- $\varepsilon = 0.2$: imperfect substitutability between C-bank and S-bank liquidity.
- $\pi_B = 85\%$: bailout probability for S-bank default in the benchmark.

Interpretation. The calibration gives S-banks a meaningful but weaker role in liquidity production than insured deposits. It also gives them substantial implicit support through a high bailout probability, which is important for matching high shadow-bank leverage.

6.2 Table 2: Commercial–shadow bank debt price spread regressions

Read. The dependent variable is the quarterly spread between the price of C-bank debt and the price of S-bank debt. The regressors are S-bank debt/GDP, C-bank debt/GDP, consumption/GDP, the federal funds rate, and in one specification the VIX.

Key numbers.

- log S-bank debt/GDP coefficient: **-0.19%** without VIX and **-0.14%** with VIX.
- log C-bank debt/GDP coefficient: **0.50%** without VIX and **0.44%** with VIX.
- adjusted R^2 : about **0.77–0.78**.

Interpretation. Increasing the supply of one type of liquidity affects the relative price of the other. This supports the model’s assumption that C-bank and S-bank debt are imperfect substitutes.

TABLE 1 Calibration					
Panel A: Parameters with a one-to-one mapping to the data					
	Value	Description	Target		
Production					
ρ_Y	0.599	Non-bank GDP autocorr.	AC(NIPA GDP - Bank dep. GDP)		
σ^F	0.87%	Non-bank GDP volatility	Vol(NIPA GDP - Bank GDP)		
δ_K	2.5%	Depreciation	Quarterly depreciation NIPA		
η	0.667	Labour share	NIPA labour share		
$\sigma_{\epsilon,C}$	12.1%	SD C-bank idiosync. shocks	vol(equity payout C-banks)		
$\sigma_{\epsilon,S}$	25.4%	SD S-bank idiosync. shocks	vol(equity payout S-banks)		
Bank regulation					
θ	10%	Capital requirement	mean(Tier-1 Eq/Assets)		
κ	0.142%	Deposit insurance fee	FDIC report (see table notes)		
Runs					
Z	$26\% \times Z$	Real asset foreclosure discount	Campbell, Giglio, and Pathak (2011)		
π^R	[0, 0.333]	Fraction of households run	Covitz, Liang, and Suarez (2013)		
Prob _{run}	[0.963 0.0375]	[I]Unconditional run probability	[I]4.76%		
	[0.75 0.25]	Average bank run length	1.3 quarters		
Panel B: Parameters that jointly match moments in the data					
	Values	Description	Target	Data (%)	Model (%)
Bank technology					
ν^Z	0.23	Scales bank output	Real GDP per capita share	22.3	22.3
σ^Z	1.74%	Bank dep. output volatility	Vol(Bank output growth)	2.93	3.02
ϕ_I	0.3	Investment adj. cost	Vol(Investment)	2.65	1.04
ϕ_K	0.011	Capital growth adj. cost	Vol(C-bank asset growth)	0.50	0.46
Bank default					
δ_S	0.39	Default penalty S-banks	S-bank bond default bond default rate	0.28	0.30
S-bank bond default					
δ_C	0.204	Default penalty C-banks	Loan net charge-offs	0.23	0.23
ξ_S	20.5%	Bankruptcy cost S-banks	Unsecured debt recov. Moody's	38.1	38.2
ξ_C	35.2%	Bankruptcy cost C-banks	Sec. debt recov. & FDIC resol. costs	48.1	48.1
π_B	85%	Bailout probability	Shadow bank leverage	87.0	83.2
Liquidity preferences					
β	0.993	Discount rate	C-bank interest rate on deposits	0.36	0.39
α	0.33	CES weight S-bank debt	S-bank share Gallin (2015)	34.0	33.7
ψ	0.0072	Liq. preference weight	Liq. premium BDG2019	0.21	0.17
γ_H	1.6	Liq. preference curvature	Reg. coefficient on AS	-0.19	-0.14
ϵ	0.2	Liq. type elasticity	Reg. coefficient on AC	0.50	0.68
Run					
		Capital dep.	15.1	15.2	
		Avg. haircut Gorton and Metrick (2009)			

Notes: This table lists the parameters of our model. See discussion in Section 4.2 on data counterparts of model variables and liquidity demand parameters. All other parameters are discussed in [Supplementary Appendix D.2](#). The data sources for this table are from NIPA, Flow of Funds, FRED (<https://fred.stlouisfed.org/>), Compustat/CRSP and the FR-Y-9C. Sample period is from 1999Q1 to 2019Q4. The link to the FDIC report is <https://www.fdic.gov/about/strategic/report/2016annualreport/ar16section3.pdf>. The volatility of equity payout is the time series average from Compustat/CRSP of the cross-sectional volatility of equity payout (dividend + net repurchases) per share for commercial banks and shadow banks (see [Supplementary Appendix](#) for definition using SIC codes). Using the same definition and data source, shadow bank leverage is the value weighted debt to asset ratio. BDG2019 stands for Van Binsbergen *et al.* (2021).

TABLE 2
Commercial—shadow bank debt price spread regressions

	(1)	(2)
log(S-bank debt/GDP)	-0.19% (-1.86)	-0.14% (-1.13)
log(C-bank debt/GDP)	0.50% (1.64)	0.44% (1.45)
Federal funds rate	54.89% (10.02)	54.61% (10.66)
log(C/GDP)	0.57% (0.26)	-0.24% (-0.11)
VIX		-0.12% (-1.39)
Adj. R^2	0.77	0.78

Notes: The sample is 1999Q1–2019Q4. The dependent variable is the quarterly spread between the price on commercial bank debt, defined as the inverse of the interest rate paid on deposits, and the price on shadow bank debt, defined as the 3-month AA-financial commercial paper price. The interest rate paid on deposits is calculated as the aggregate amount of interest rate expense on deposits divided by the aggregate amount of deposits. The data come from FR-Y-9C reports (bank holding companies). We download the 3-month AA-financial commercial paper rate from FRED (i.e. Federal Reserve Economic Data maintained by the Federal Reserve bank of St. Louis). We download the Fed Funds rate from FRED. Shadow bank debt is the sum of REPO claims held by shadow banks (Flow of Funds table L.207), total money market mutual fund assets (equals liabilities) (table L.206), and commercial paper held by the domestic financial sector less depository institutions (table L.209). Commercial bank debt is set to aggregate bank deposits (FR-Y-9C). We download the real consumption series from FRED. All quantities are normalized by GDP and logged. The VIX is the daily CBOE Volatility Index downloaded from FRED, converted into a quarterly average. We use the HAC estimator to compute heteroscedasticity and autocorrelation consistent standard errors. We report the resulting t-statistics in brackets. Both regressions include a constant term.

6.3 Table 3: Properties of the dynamic model

Read. Table 3 compares untargeted volatility and business-cycle correlations in the data and model.

Key results.

- Consumption volatility is close: **0.38%** in data and **0.39%** in model.
- Consumption and investment are strongly procyclical in both data and model.
- C-bank market leverage is countercyclical in both data and model.
- The C-bank liquidity benefit is countercyclical in both data and model.

Interpretation. The model is not designed only to match calibration targets. It also captures several relevant financial-cycle patterns, though it misses the level of C-bank leverage volatility and produces countercyclical interest rates.

6.4 Figure 1: The effect of bank runs

Read. Figure 1 compares impulse responses to a productivity shock alone with impulse responses to a productivity shock plus an S-bank run shock.

Key result. Adding a bank run makes the recession much worse. Output, consumption, investment, and liquidity services fall more sharply. Deadweight losses rise by about **3.5 percentage points**.

Interpretation. Runs matter because S-banks must liquidate assets at fire-sale values. Shadow banking is fragile not merely because it defaults, but because runs force inefficient liquidation and reduce aggregate liquidity.

TABLE 3
Properties of the dynamic model

	Data	Model
Volatility		
Consumption	0.38%	0.39%
Mkt Lev C	1.48%	0.10%
Book Lev C	0.29%	0.09%
Mkt Lev S	2.22%	3.51%
Book Lev S	3.41%	3.51%
Debt C/Debt S	0.52	0.20
Business cycle correlation		
Consumption	0.89	0.72
Investment	0.92	0.95
Debt S Share	0.43	0.03
Debt S (lag GDP)	0.26	0.15
Book Lev C	0.16	0.43
Mkt Lev C	-0.21	-0.42
Book Lev S	0.04	-0.00
Mkt Lev S	-0.31	-0.02
C-bank liquidity benefit	-0.26	-0.52
Yield C	0.74	-0.85
Yield S	0.82	-0.79
rf. rate	0.70	-0.85
Spread S-C	0.51	0.12

Notes: This table presents untargeted moments in the model and compares them to their data counterpart. The sample for the data is 1999 Q1–2019 Q4. The volatility is the standard deviation of the HP-filtered variable. The GDP correlation is the correlation of HP-filtered business cycle component of GDP with the HP-filtered business cycle component of the variable of interest. See [Supplementary Appendix D.4](#) for the description of the data.

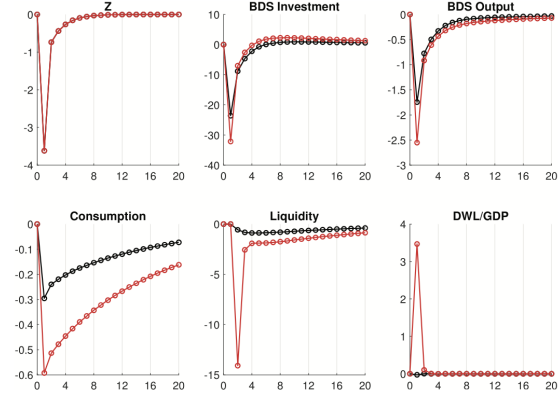


FIGURE 1
The effect of bank runs on the economy

This figure presents the impulse response functions to a productivity shock (in black) and a productivity shock together with a run shock (red). The x-axis denotes quarters. The shocks occur in the first quarter. The y-axis denotes percentage deviations from the stationary equilibrium for all plots but bottom right (DWL/GDP), which shows the difference to the stationary equilibrium in percentage points.

6.5 Table 4: Effect of higher capital requirements

Read. Table 4 simulates the model under alternative C-bank capital requirements, holding all other parameters fixed. The benchmark is $\theta = 10\%$; the table considers requirements up to 30%.

Key benchmark moments.

- S-bank debt share: **31.95%**.
- S-bank capital share: **33.68%**.
- S-bank leverage: **83.18%**.
- C-bank leverage: **89.95%**.
- S-bank default: **0.30%**.
- C-bank default: **0.23%**.

At the optimal 16% capital requirement.

- S-bank debt share rises by **4.61%** relative to benchmark.
- S-bank leverage rises by **0.43%**.
- C-bank leverage falls by **6.67%**.
- S-bank default rises by **7.39%**.
- C-bank default falls by **89.38%**.
- liquidity services fall by **4.22%**.
- consumption rises by **0.086%**.
- welfare rises by **0.0540%**.

Interpretation. Tighter regulation causes leakage to shadow banks and makes S-banks somewhat riskier. But the fall in C-bank risk is much larger, so aggregate consumption and welfare rise.

TABLE 4
Effect of higher capital requirements

	Base	13%	14%	15%	16%	17%	20%	30%
Capital and debt								
1. Capital	3.15	0.16%	0.23%	0.30%	0.38%	0.46%	0.72%	1.64%
2. Debt share S	31.95%	2.71%	3.39%	4.01%	4.61%	5.18%	6.91%	13.79%
3. Capital share S	33.68%	0.26%	0.09%	-0.15%	-0.43%	-0.73%	-1.73%	-4.79%
4. Capital S	1.06	0.42%	0.32%	0.16%	-0.05%	-0.28%	-1.02%	-3.23%
5. Leverage S	83.18%	0.18%	0.26%	0.34%	0.43%	0.52%	0.80%	1.80%
6. Leverage C	89.95%	-3.33%	-4.44%	-5.56%	-6.67%	-7.78%	-11.12%	-22.22%
7. Early Liquidation	0.00	0.25%	0.36%	0.47%	0.59%	0.72%	1.10%	2.51%
Prices								
8. Deposit rate S	0.45%	-0.66%	-0.96%	-1.28%	-1.61%	-1.96%	-3.05%	-6.80%
9. Deposit rate C	0.39%	-3.69%	-4.86%	-6.01%	-7.17%	-8.35%	-12.04%	-26.83%
10. Convenience Yield S	0.28%	1.39%	2.00%	2.65%	3.33%	4.04%	6.26%	14.33%
11. Convenience Yield C	0.31%	4.68%	6.14%	7.60%	9.06%	10.54%	15.17%	33.97%
Welfare								
12. Default S	0.30%	3.05%	4.40%	5.85%	7.39%	9.00%	14.12%	34.08%
13. Default C	0.23%	-5.11%	-76.16%	-83.96%	-89.38%	-93.09%	-98.28%	-100.00%
14. GDP	1.29	0.01%	0.02%	0.02%	0.03%	0.03%	0.05%	0.12%
15. Liquidity Services	1.48	-2.16%	-2.85%	-3.54%	-4.22%	-4.90%	-6.96%	-14.09%
16. Consumption	1.21	0.062%	0.073%	0.081%	0.086%	0.090%	0.098%	0.107%
17. Vol(Liquidity Services)	0.03	-3.16%	-4.26%	-5.39%	-6.53%	-7.66%	-11.08%	-22.31%
18. Vol(Consumption)	0.00	0.31%	0.34%	0.36%	0.36%	0.36%	0.31%	0.07%
19. Welfare	0.0460%	0.0511%	0.0535%	0.0540%	0.0527%	0.0435%	0.0053%	

This table shows the moments of the simulated model for different values of C-banks' capital requirement θ . The "Base" titled column shows the moments for the benchmark calibration. All other columns show the percentage change relative to the benchmark moment.

TABLE 5
Higher capital requirements without demand effect ($\gamma_H = 0$)

	(1) C-bank only				(2) Simple model				(3) No demand effect			
	Base	14%	15%	16%	Base	14%	15%	16%	Base	14%	15%	16%
Capital and debt												
1. Capital	3.17	-0.64%	-0.76%	-0.89%	3.20	-0.38%	-0.45%	-0.52%	3.17	-0.40%	-0.47%	-0.54%
2. Debt share S	-	-	-	-	39.73%	4.37%	5.24%	6.08%	31.75%	4.15%	4.95%	5.72%
3. Capital share S	-	-	-	-	41.83%	1.60%	1.77%	1.89%	31.41%	1.20%	1.34%	1.53%
4. Leverage S	-	-	-	-	82.47%	-0.10%	-0.12%	-0.15%	83.47%	-0.42%	-0.50%	-0.58%
5. Leverage C	89.99%	-4.44%	-5.55%	-6.66%	89.99%	-4.45%	-5.56%	-6.67%	89.94%	-4.43%	-5.54%	-6.65%
6. Run liq	-	-	-	-	-	-	-	-	0.00	-0.62%	-0.73%	-0.84%
Prices												
7. Deposit rate S	-	-	-	-	0.46%	1.92%	2.30%	2.68%	0.43%	1.69%	2.01%	2.31%
8. Deposit rate C	0.37%	0.09%	0.14%	0.22%	0.35%	-1.93%	-2.31%	-2.68%	0.38%	-1.34%	-1.60%	-1.85%
9. Conv. Yield S	-	-	-	-	0.24%	-3.76%	-4.50%	-5.21%	0.29%	-3.15%	-3.75%	-4.33%
10. Conv. Yield C	0.33%	0.20%	0.22%	0.23%	0.35%	1.94%	2.32%	2.68%	0.32%	1.58%	1.88%	2.15%
Welfare												
11. Default S	-	-	-	-	0.01%	-4.61%	-4.81%	-5.57%	0.33%	-6.62%	-7.87%	-9.05%
12. Default C	0.23%	-76.20%	-83.97%	-89.37%	0.23%	-76.26%	-84.02%	-89.42%	0.23%	-76.14%	-83.95%	-89.38%
13. GDP	1.29	-0.05%	-0.06%	-0.07%	1.29	-0.03%	-0.03%	-0.04%	1.29	-0.03%	-0.04%	-0.04%
14. Liq. Serv	2.85	-5.06%	-6.28%	-7.59%	1.47	-3.77%	-4.64%	-5.50%	1.49	-3.72%	-4.60%	-5.48%
15. Cons.	1.21	0.10%	0.11%	0.12%	1.21	0.06%	0.07%	0.07%	1.21	0.07%	0.07%	0.08%
16. Vol(Liq. Serv)	0.06	-3.98%	-5.02%	-6.04%	0.03	-3.76%	-4.65%	-5.52%	0.05	0.43%	0.30%	0.12%
17. Vol(Cons)	0.00	0.31%	0.27%	0.19%	0.00	0.61%	0.69%	0.76%	0.00	0.00%	0.02%	0.04%
18. Welfare	0.00	0.070%	0.071%	0.069%	0.00	0.035%	0.034%	0.031%	0.00	0.066%	0.047%	0.045%

This table presents moments of the simulated model for different model versions and different values of C-banks' capital requirements θ . All cases are calculated for $\gamma_H = 0$, which means that there is no demand effect. The "Base" column presents model moments (means and volatilities) with the benchmark capital requirement level. The other columns present results as percentage deviations from the "Base" level for increasing the capital requirement to the "Simple" percentage. The columns headed with "C-Bank Only" presents results for increases in the capital requirement in a model with only regulated commercial banks. The columns headed with "Simple Model" present results with the parameterization that mimics the model described in Section 3. Finally, the columns headed with "No Demand Effect" show the results for the full quantitative model about a demand for deposits, i.e., $\gamma_H = 0$. Shadow banks are abbreviated with "S" and commercial banks with "C". "Run liq" stands for early liquidations during a run, "Conv" for convenience, "Cons" for consumption, and "Liq. Serv" for liquidity services.

6.6 Table 5: Higher capital requirements without demand effect

Read. Table 5 sets $\gamma_H = 0$ to remove the demand effect and studies three simplified model variants: C-bank only, simple model with S-banks, and full model without demand effect.

Key lesson.

- Without the demand effect, higher capital requirements make S-banks safer.
- In the full model without demand effect, raising θ to 15% lowers S-bank leverage by about **0.50%** and lowers S-bank default by about **7.87%**.
- In the full benchmark with demand effect, raising θ to 16% raises S-bank leverage and S-bank default.

Interpretation. Table 5 is the cleanest decomposition of the paper. It shows that the sign of S-bank risk-taking depends on whether the demand effect is active. The competition effect alone makes S-banks safer; the demand effect can reverse this.

6.7 Figure 2: Recovery from the 2008 financial crisis

Read. Figure 2 simulates a financial crisis and post-crisis reform. The pre-crisis model has lower capital requirements, higher S-bank bailout probability, and underestimation of run risk. After the crisis, capital requirements rise, the S-bank bailout probability falls, and agents correctly recognize run risk.

Key numbers and patterns.

- The pre-crisis S-bank debt share is about **42%**, close to the data.
- A run shock causes a sharp contraction in bank-dependent output.
- Consumption falls by close to **1%** and liquidity services by more than **15%**.
- Raising capital requirements alone would slightly increase the S-bank share.
- But the post-crisis fall in bailout expectations and the recognition of run risk reduce the S-bank share by about **10 percentage points**.

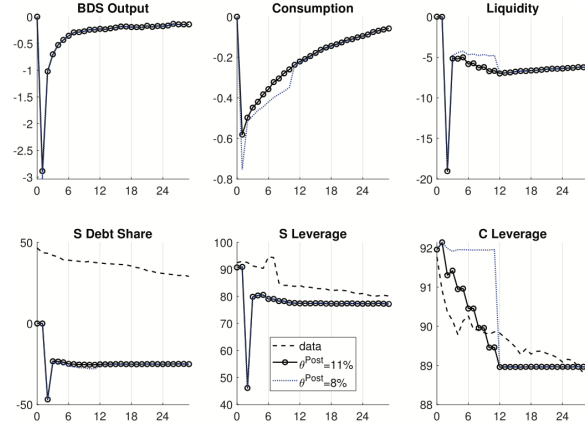


FIGURE 2
Recovery from 2008 financial crisis in model simulations

This figure plots the time path of several model variables in a simulation of the 2008 financial crisis. The y-axis denotes percentage deviations from the initial state for the four panels in the top row, and percent in the four bottom panels. The solid black lines with circles plot the baseline simulation described in the text that raises capital requirements to 11% post-crisis. The dotted blue line includes the same parameter changes as the black line, except the increase in capital requirements. For the bottom three panels, the dashed line plots data counterparts to the model variables as described in [Supplementary Appendix D.5](#).

Interpretation. The model supports the idea that higher bank capital requirements push activity toward shadow banks, but this effect is quantitatively dominated after 2008 by the decline in implicit shadow-bank guarantees and the recognition of run risk.

7 Identification, credibility, and limitations

7.1 What identifies what?

- **Technology parameters** are disciplined by output, investment, depreciation, labor share, and bank output volatility.
- **Default and recovery parameters** are disciplined by C-bank charge-offs, S-bank bond defaults, and debt recovery data.
- **Run parameters** are disciplined by evidence on run frequencies, run lengths, and haircuts.
- **Bailout probability** is disciplined by S-bank leverage. Without a substantial implicit guarantee, the model cannot match high S-bank leverage.
- **Liquidity preference parameters** are disciplined by C-bank deposit rates, shadow-bank shares, liquidity premium estimates, and the spread regressions in Table 2.

7.2 Why the exercise is credible

- The paper targets moments that are close to the mechanisms of interest: leverage, default, recoveries, deposit rates, liquidity premia, and shadow-bank shares.
- The key mechanism is disciplined by relative quantities and prices of safe and shadow-bank debt rather than assumed qualitatively.
- The model is checked against untargeted dynamic moments, not only calibration targets.
- The paper provides analytical intuition through a simple model and then shows how the same forces operate quantitatively.

- The table decompositions show which conclusions depend on the demand effect and which come from the competition effect.

7.3 Important limitations

- **No asset-side heterogeneity.** C-banks and S-banks hold the same assets. This isolates liability-side mechanisms but abstracts from specialization, screening, origination, securitization, and regulatory arbitrage on the asset side.
- **Reduced-form liquidity demand.** Money-like services are modeled through utility from debt holdings. This is standard and tractable, but the microfoundation is not the focus.
- **Bailouts are exogenous.** The probability of S-bank bailouts is calibrated rather than derived from political economy or optimal government intervention.
- **Tightly binding C-bank constraint.** The model misses some observed C-bank leverage volatility because real banks hold capital buffers.
- **No monetary policy channel.** The model produces countercyclical rates, a known limitation of CRRA models without richer monetary-policy or preference structure.

Bottom line. The paper is strongest as a quantitative mechanism paper: it gives a disciplined way to think about regulatory leakage and shadow banking. It is less suited for answering questions that require detailed asset-side heterogeneity or endogenous political decisions about bailouts.

8 Short summary

- The paper builds a quantitative general equilibrium model with insured commercial banks and uninsured/runnable shadow banks.
- Both bank types produce credit and money-like liabilities; households value their liabilities as imperfectly substitutable liquidity services.
- Deposit insurance gives C-banks a competitive advantage, making C-banks too large and pushing S-banks to lever up because S-bank liquidity is scarce.
- A higher C-bank capital requirement weakens this competitive advantage and shifts activity toward S-banks.
- Whether S-bank leverage rises or falls depends on two forces:
 - competition effect: lowers S-bank leverage;
 - demand effect: raises liquidity premia and increases S-bank leverage.
- In the calibrated model, the demand effect dominates for S-bank leverage, so tighter C-bank regulation makes S-banks larger and slightly riskier.
- The overall financial system nevertheless becomes safer because the reduction in C-bank default risk is much larger.
- The welfare-maximizing C-bank capital requirement is around **16%**.

- The post-crisis decline in shadow banking can be reconciled with the model once one accounts for lower implicit bailout probabilities and the recognition of run risk.

9 Conclusion

Core conclusion. Capital regulation of commercial banks does create substitution toward shadow banking. However, in the calibrated model this leakage does not overturn the benefits of regulation. A system with more shadow banking can still be safer if commercial-bank risk falls enough.

Economic intuition. Deposit insurance subsidizes C-bank debt and distorts competition. Shadow banks respond to the scarcity of their liquidity services by taking leverage risk. Capital requirements reduce the C-bank subsidy advantage but also raise aggregate liquidity premia. The balance of these two forces determines S-bank risk-taking.

Policy takeaway. Regulators should not evaluate capital requirements by asking only whether activity migrates to shadow banks. The key question is whether migration increases or decreases total fragility after accounting for liquidity demand, implicit guarantees, and competition between bank types. In this model, the optimal policy is tighter C-bank capital regulation, with an optimal requirement of about **16%**.